Ayush Madhav Kumar

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Education

University of Michigan, College of Engineering **Aug 2025 – Dec 2028**
B.S.E. Computer Science, Minor in Mathematics; GPA 3.60/4.00 *Ann Arbor, MI*
– **Coursework:** Data Structures and Algorithms, Computer Organization, Discrete Mathematics, Linear Algebra

Douglas Community School **Mar 2023 – Jun 2025**
Leaving Certificate: Mathematics, Computer Science, Physics, Chemistry, Applied Mathematics *Cork, Ireland*

Global Schools Group **Jun 2021 – Mar 2023**
High School Diploma, Mathematics and Science *India*

Research Experience

NSF POSE Research, University of Michigan **May 2026 – Present**
Undergraduate Student Researcher *Ann Arbor, MI*
– Building the benchmark suites for Jac's experimental GPU walker compiler under Professors Jason Mars and Lingjia Tang, on an NSF Phase I grant ($300K) growing the open source ecosystem around Jac
– Designed a layout benchmark that holds the checked LLVM O2 PTX fixed and changes only node order in memory, comparing random and access-order placement against a shared-memory tiled CUDA reference, timed in Nsight Systems on an RTX 3090
– Access-order placement made the same kernel 1.11×, 1.14×, and 1.75× faster on the three larger matrices, consistent across three seeds, but 6–12% slower on the two smallest, supporting a size threshold for reordering; the Jac kernel still trailed tiled CUDA by 5–34×
– Wrote six graph application benchmarks (DAG dependency analysis, device hierarchies, network outage propagation, fraud screening, sparse rows, and sequences) timing the full `run_walkers` call, including packing, transfer, kernel, copyback, and result publication
– Shared-graph workloads ran 9–94× faster than serial Jac while private-subgraph workloads gained only 1.6–3.4×; showed that speedup tracks average visits per node and pointed to per-node Python graph packing as the cheapest next fix
– Checked every CPU and GPU output against independent oracles in all 19 application cases over 3 warmup and 15 measured rounds in randomized arm order, and wrote the run guide and results report
– Earlier built Marketing Agent Studio, an autonomous content agent in 55 Jac modules with 166 tests, and NightShift, an unattended pipeline that audits the 365K-line Jac compiler nightly and opens draft PRs behind a fail-closed CI mirror

Michigan CLAWS, NASA SUITS Challenge **Aug 2025 – Present**
Artificial Intelligence Research *Ann Arbor, MI*
– Co-author, ACM UIST 2026 poster on the GEMINI augmented-reality EVA assistant (see Publications)
– Developed the autonomous lunar EVA navigation research stack in the DUST simulator: A* and Jump Point Search planning over a 3-layer occupancy grid, 17-ray LiDAR fusion, PD heading control with curvature lookahead, and RSSI trilateration via inverse log-distance modeling, evaluated to 96.8% waypoint accuracy
– Trained and compared two obstacle detection architectures, a temporal ResidualBeamCNN and a BEV cross-attention transformer with learnable query embeddings, served in real time behind a hot-swappable multi-backend inference framework
– Built the speech interface studied in the poster work, streaming 16 kHz PCM over WebSocket into VAD, faster-whisper, and a neural intent classifier that routes to 50+ HoloLens AR scene actions at 200ms end to end
– Designed CORVUS, a dual-inference architecture running TinyBERT and DistilBERT on a Jetson Orin Nano through ONNX Runtime for sub-350ms deterministic responses under varying network conditions

Industry Experience

Jaseci Labs **May 2026 – Aug 2026**

Software Engineering Intern *Ann Arbor, MI*

- Fine-tuned Qwen3-Coder 30B into a coding agent for Jac, an object-spatial superset of Python where general purpose models have no usable priors, reaching 94% compile-pass on a held-out benchmark
- Generated 300k+ training examples through a multi-recipe synthetic data pipeline spanning Python to Jac parallel translation, adversarial DPO negatives, evol-instruct, and self-distillation, with every example gated behind the Jac compiler and unit tests before entering the corpus
- Designed a four-stage training curriculum across core SFT, debugging and reasoning, DPO alignment, and multi-turn agentic data, cutting the model's fallback to Python patterns by 82% and locking in idiomatic walker, node, and edge constructs
- Built training data spanning natural language to Jac generation, compiler error debugging, Python to Jac conversion, and MCP-driven subagent orchestration for multi-step agentic tasks
- Targeted a roughly 9B parameter model that writes Jac better than frontier-scale models while remaining small enough to run at the edge, evaluated against a compiler-verified held-out set
- Contributed to Jac ML Studio, a local ML workbench pairing an MLX and LoRA training pipeline with a pure-Jac web application serving both endpoints and a React-in-Jac client from one codebase, with job tracking and eval dashboards

Cactus (YC S25) **Jan 2026 – Present**

Open Source Contributor *Remote*

- Contribute to an open source mobile inference engine for smartphones and low power devices, working in the C++ engine core
- Fixed image completions hanging on both Metal and CPU: the encoder reported full context length, so a shape rewrite mismatch split a 1,043-token prompt into roughly 1,000 prefill chunks instead of 9, and a missing manifest key silently routed LFM2-VL bundles down the wrong preprocessing path
- Fixed byte-fallback decoding in metaspace BPE tokenizers, where unmapped codepoints surfaced as literal hex pieces instead of correct multi-byte characters
- Fixed SentencePiece vocabulary loading for tokens containing tabs, which previously threw during initialization or silently truncated the token
- Documented custom vocabulary and vocabulary boost options for transcription across the engine docs and the Python, Swift, Kotlin, and Flutter SDK READMEs
- Verified each fix against the model and hardware it affected, reporting before and after behavior on quantized vision-language builds on Apple silicon across both the Metal and CPU backends

CareTether (Patch Summer Accelerator '25) **Jul 2025 – Aug 2025**

Co-Founder; Frontend, Sales and Outreach *Dublin, Ireland*

- Co-founded a wearable and companion app for seniors with cognitive difficulties, built on site over an 8-week residential entrepreneurship accelerator
- Built the mobile front end in React Native and ran a pilot with 5+ families
- Reached 20,000+ people through social campaigns, and pitched distribution partnerships to NGOs and care homes
- Took the product from concept to working pilot inside the cohort and presented it to investors at the end of the programme

GIIT Solutions **Jun 2024 – Jul 2024, Feb 2025**

Associate Software Intern *Cork, Ireland*

- Built and maintained CI/CD deployment pipelines on AWS using GitHub Actions
- Wrote Python tooling to automate backend release processes and reduce manual overhead
- Worked with senior engineers on cloud resource allocation and cost efficiency
- Deployed cloud-native services across the team's AWS estate and helped keep deployment paths repeatable across environments

Club Experience

Michigan CLAWS, NASA SUITS Challenge **Aug 2025 – Present**
Artificial Intelligence Lead; previously AI Software Developer *Ann Arbor, MI*

- Lead the AI subteam of 6, setting technical direction and dividing work across navigation, perception, and voice; team poster accepted to ACM UIST 2026 (Detroit, November 2026)
- Wrote the autonomous lunar EVA navigation stack in the DUST simulator: A* and Jump Point Search planning over a 3-layer occupancy grid, 17-ray LiDAR fusion, PD heading control with curvature lookahead, stuck detection, and RSSI trilateration via inverse log-distance modeling, reaching 96.8% waypoint accuracy
- Trained the obstacle detection models, a temporal ResidualBeamCNN and a BEV cross-attention transformer with learnable query embeddings, served in real time behind a hot-swappable multi-backend inference framework
- Built the voice pipeline at 200ms end to end, streaming 16 kHz PCM over WebSocket into VAD, faster-whisper, and a neural intent classifier that routes to 50+ HoloLens AR scene actions
- Designed CORVUS, a dual-inference architecture for Project GEMINI, running TinyBERT and DistilBERT on a Jetson Orin Nano through ONNX Runtime for sub-350ms deterministic responses, with DSPy orchestration, Pydantic-validated command routing, and fail-safe fallbacks across network conditions
- Built an adaptive edge and cloud routing layer that dispatches inference by query complexity, latency budget, and network state, so EVA autonomy degrades gracefully instead of failing when connectivity drops

WolvSec, University of Michigan Cybersecurity Club **Aug 2025 – Present**
Vice President; previously Developer and Member *Ann Arbor, MI*

- Elected Vice President in Mar 2026 after serving as Developer; lead weekly training sessions and competition preparation for Michigan's largest cybersecurity club
- Represented Michigan in 12 CTFs, finishing first twice and solving 50+ challenges across web and API exploitation, reverse engineering, packet forensics, and cryptography
- Authored 12 WolvCTF 2026 challenge submissions from beginner to hard: 5 binary exploitation, 5 web and application security, 1 cryptography, and 1 forensics
- Produced player handouts spanning stripped ELF binaries, containerized services, static cryptography artifacts, and a FAT12 disk image, with build tooling and author solvers or writeups where appropriate
- Built jailed KCTF services and solver-backed healthchecks for hosted submissions, validating intended solves through local runs before event review
- Selected hard submissions received source-blind solves or playtests to check that players could follow the intended path without access to author source

Publications

GEMINI: A Multi-Modal Augmented Reality Interface for Astronaut Guidance During Lunar Surface Operations. In *Adjunct Proceedings of the 39th Annual* ***ACM Symposium on User Interface Software and Technology (UIST Adjunct '26)***, Detroit, MI. **Poster; presenting November 2026.**

- AR heads-up display on the Microsoft HoloLens 2 that guides astronauts through a full lunar EVA: pressurized rover egress and ingress, terrain navigation, suit telemetry monitoring, and in-field lunar terrain vehicle repair
- Voice is the primary input, with eye gaze and a leg-mounted button as a redundant path for gloved, suit-constrained hands; a Flask backend streams NASA suit, DCU, UIA, and rover telemetry to the Unity HUD over Socket.IO
- Assistant (Corvus) uses a fine-tuned all-MiniLM-L6-v2 model for intent classification and entity extraction, chosen over a generative model to keep safety-critical commands deterministic and low latency
- **My contribution:** led the AI subteam and built the speech interface, streaming 16 kHz PCM over WebSocket into VAD, faster-whisper, and the intent classifier, routing to 50+ HoloLens scene actions at 200ms end to end
- Evaluated at the NASA JSC Rock Yard at night under simulated lunar lighting with a NASA AR systems engineer and an astronaut, using think-aloud, SUS, and NASA-TLX; the engineer rated the system 4/5 overall and 7/10 for navigation confidence

Hackathons and Awards

ConceptPilot | ***1st Overall, Education Track, Red Bull x Kappa Theta Pi Hackathon, Mar 2026***

- Turns raw exam results and a concept map into readiness insights, so an instructor sees which ideas a cohort has actually absorbed rather than a column of scores
- Clusters struggling students by the concepts they miss together, then traces each cluster back to the root-cause prerequisite rather than the topic where the failure surfaced
- Presents the result as an interactive prerequisite graph showing dependencies and gaps at a glance, so remediation can target upstream causes

SaveSurance | ***Winner, ElevenLabs Track, MLH RevolutionUC, Mar 2026***

- Multimodal medical bill parser that takes a photographed or uploaded bill and returns structured line items, savings signals, and financial-aid eligibility screening
- Dual-path backend so parsing degrades to a second route rather than failing outright on bills the primary path cannot read
- Generates editable PDF appeal packets from the parsed bill, and runs a live call assistant over WebSocket so a patient can be coached through a call with a billing department
- Team project, deployed and live at savesurance.tech

GhostWatch | ***2nd Overall and cash prize, Jaseci Hackathon, Apr 2026***

- Security and pull request review built in Jac, aimed at open source maintainers reviewing contributions from people they do not know
- Verified GitHub webhooks fire graph walkers that map the codebase as a graph, then score each change by dependency impact and blast radius rather than by diff size
- Surfaces the resulting risk in a web control room, so a maintainer sees which pull requests touch load-bearing code before reading any of them

SiteSense | ***Winner, Audited AI Track, Anthropic Claude Hackathon, Nov 2025***

- Browser extension that reads the privacy policy, cookie notice, and extension permissions of a site the user is already on, rather than asking them to find and read the documents themselves
- Assigns a privacy risk score with Claude and surfaces only the warnings that change how the user's data is handled, so the output is a decision rather than a summary
- Deployed and live at your-site-sense.vercel.app

PreReq | ***Top 5, 24-Hour AI and Business Hackathon, Feb 2026***

- Concept-readiness platform built for both sides of a classroom from one upload of exam scores and a concept graph
- Instructors get a class heatmap with root-cause traces showing where a gap originates; students get a personal report with a study plan ordered by prerequisite rather than by syllabus

Selected Projects

OSINT Geospatial Intelligence Platform | ***TypeScript, React, CesiumJS, Node.js, SQLite***

- Full-stack intelligence platform ingesting live public feeds, military ADS-B aircraft, USGS seismic events, Safecast radiation sensors, ISS orbital telemetry, and airport airspace, and streaming them onto an interactive 3D globe with a tactical command-center interface
- Wrote a declarative ingestion engine that onboards any new feed as a single YAML file with no engine changes, with pluggable JSON, GeoJSON, XML, and CSV parsers, a dot and bracket path resolver, derived fields, filter rules, and HTTP polling with configurable intervals, timeouts, and exponential-backoff retry
- Designed the SQLite schema of sources, entities, and observations with WAL journaling, cascade deletes, and category indexes; eliminated a 372K-row duplication bug across 31K entities and 780K+ observations with a deterministic observation ID and a unique natural-key index, and capped per-entity track retention to bound growth
- Built the real-time telemetry channel with a category-balanced initial snapshot so every layer renders on first paint, delta-only broadcasts that fire only when an entity is new or has moved, and a heartbeat that terminates dead sockets
- Kept tens of thousands of live markers interactive on integrated GPUs by rendering through a single BillboardCollection rather than one entity per target, enabling on-demand rendering, and adding a level-of-detail toggle with a live FPS readout
- Wrote 142 tests across an npm workspaces monorepo, covering parsers, field mapping, the fetcher and scheduler, socket lifecycle, REST routes, and UI components, behind an OpenAPI 3.0 documented Express API

Internship Tracker | ***Python, FastAPI, PostgreSQL, React, Amazon Bedrock, AWS***

- Deployed full-stack tracker that scrapes SimplifyJobs, speedyapply, and the Direct Consideration newsletter on a schedule, keeping only the Summer 2027 US software and AI/ML internships open to an international Bachelor's student who needs sponsorship, about 1,460 live listings
- Vets postings that lack eligibility flags by pulling their text from the Greenhouse, Lever, Ashby, and Workday APIs, applying keyword rules, and sending only the undecided ones to Claude Haiku on Amazon Bedrock with adaptive retries; merges duplicate roles across sources by normalizing URLs and stripping tracking parameters
- Tracks applications automatically: forwarded recruiter emails reach an AgentMail webhook verified with Svix HMAC signatures, and Claude classifies each one to move the application to OA, Interview, Offer, or Rejected
- Runs on a single EC2 Graviton instance provisioned with CloudFormation, using Docker Compose, Caddy auto-HTTPS with a password gate, an IAM instance role in place of stored keys, and daily pg_dump backups to S3 with 30-day retention

SessionSwitch | ***Swift 6, SwiftUI, AppKit, Carbon, SwiftPM***

- macOS app for managing Claude Code CLI sessions spread across terminal windows and IDEs, so model and reasoning effort can be switched without typing commands into each session separately
- Uses AppKit overlay badges to label live sessions in place over their terminal windows, and Carbon hotkeys for global key handling outside the app's own focus
- Written in pure Swift 6 with no third-party dependencies, distributed through SwiftPM

Domain Portfolio Manager | ***Cloudflare Workers, Durable Objects, React, TypeScript***

- AI chat agent that manages a domain portfolio and DNS records through natural language, covering domain registration, bulk DNS updates across many zones, and queries over change history
- Runs entirely on Cloudflare's edge: Durable Objects hold per-conversation state and Workflows drive long-running multi-step operations so bulk changes survive interruption
- Deployed and live at domain-portfolio-manager-eta.vercel.app

Inferno | ***Python, Flask, RabbitMQ, MongoDB, Celery, Docker***

- Backend service that hosts ML models behind REST endpoints for model loading, inference, and health checks, supporting both TensorFlow and PyTorch model formats
- Queues inference through RabbitMQ and Celery workers for reliable processing of high-volume workloads, so request handling stays responsive under load
- Persists model outputs and user queries in MongoDB for analytics and logging, and ships containerized with Docker

2D to 3D Image Modelling | ***Python, PyTorch, Neural Radiance Fields***

- NeRF-based reconstruction tool that trains a radiance field on a set of multi-view photographs of a scene
- Renders novel views from camera positions never photographed, recovering geometry and view-dependent appearance from ordinary images with no depth sensor or scanner

ML Reads | ***Next.js, React, TypeScript, arXiv API***

- Recommends five machine learning papers from arXiv each day, with a deterministic daily selection so every reader sees the same set and the list stays finite rather than becoming an endless feed
- Parses the arXiv feed directly and surfaces titles, authors, summaries, categories, and publication dates with direct abstract and PDF links
- Deployed and live at mlreads.vercel.app

COGnitiveStudy | ***Hono, Next.js, React, TypeScript, Drizzle ORM***

- Study platform that records lectures and returns a structured summary with notes, so a student leaves a lecture with written material rather than an audio file
- Built as three layers: a Hono backend API, a typed Next.js interface, and an AI summarization stage, with Drizzle ORM over the persistence layer
- Deployed and live at cog-nitive.vercel.app

mkrl | ***FastAPI, Python, React, TypeScript, Vite***

- URL shortener with a FastAPI backend and a React and Vite front end, built stateless so it scales horizontally across instances rather than pinning traffic to one node
- Open sourced and deployed at mkrl.vercel.app

CompVis | ***Python, OpenCV, MediaPipe, NumPy, Next.js***

- Real-time face mesh detection tracking 468 facial landmarks per frame from a live camera feed
- Wraps the OpenCV and MediaPipe vision pipeline in a Next.js front end so the mesh can be viewed in a browser rather than a local window

TaskPilot | ***Django REST Framework, Python, React, TypeScript, Material UI***

- Task management platform covering the full task lifecycle, assignment, and filtering
- Django REST Framework API behind a typed React client, with Material UI for the interface layer

SMS Spam Classifier | ***Python, PyTorch, Pandas, scikit-learn, Unity Sentis***

- Spam detection model trained in PyTorch over a labelled SMS corpus with scikit-learn preprocessing
- Exported for Unity Sentis so the trained classifier runs inside a Unity runtime rather than behind a server call

Instagram CLI | ***TypeScript, React Ink, Node.js***

- Terminal client for Instagram direct messages that shows conversations without stories, reels, or the rest of the feed
- Built with React Ink, rendering a React component tree to the terminal instead of the DOM

SponsorScrape | ***Python, Playwright, AgentQL***

- Scraper that collects local restaurant data for the CLAWS sponsorship committee, replacing manual list-building for outreach
- Drives a real browser through Playwright with AgentQL queries, so extraction survives layout changes that break selector-based scrapers

Custom ESP32 Handheld | ***Embedded C, ESP32, wifi and sub-GHz radio***

- Flashed a self-written operating system onto a custom ESP32 board to work as a wifi and radio signal tool, after moving into hardware security at DEF CON 2026

Technical Skills

- **Languages:** Python, C++, C, Java, TypeScript, JavaScript, Rust, Swift, Go, C#, Ruby, PHP, SQL, Jac, Bash, HTML/CSS
- **GPU Systems:** CUDA, PTX, Nsight Systems, GPU kernel benchmarking, memory layout experiments, structure-of-arrays and CSR graph layouts, differential testing against CPU oracles
- **Machine Learning:** PyTorch, TensorFlow, JAX, HuggingFace Transformers, Unsloth, MLX, scikit-learn, LoRA/QLoRA, SFT, DPO, GRPO, RLHF, synthetic data generation, evaluation harnesses, NLP, computer vision, reinforcement learning
- **ML Systems and Inference:** vLLM, ONNX Runtime, quantization (4-bit, INT4), KV-cache optimization, on-device and edge inference, model compression, ARM SIMD kernels, RAG, vector search, agentic systems, MCP
- **Backend and APIs:** FastAPI, Flask, Django, Node.js, Express, Hono, REST, gRPC, GraphQL, WebSockets, RabbitMQ, Celery, OpenAPI 3.0
- **Infrastructure and DevOps:** Docker, Kubernetes, Linux, Amazon Web Services (EC2, Lambda, S3), Google Cloud Platform, Cloudflare Workers and Durable Objects, Terraform, Nginx, GitHub Actions, GitLab CI, launchd, serverless architecture
- **Data:** PostgreSQL, MySQL, MongoDB, Redis, SQLite, Drizzle ORM, vector databases, Pandas, NumPy, ETL pipelines, data modeling, real-time telemetry ingestion
- **Frontend and Mobile:** React, Next.js, Astro, Vite, Redux Toolkit, Tailwind CSS, Material UI, CesiumJS, Three.js, React Native, Expo, Flutter, Dart, SwiftUI, AppKit
- **Security:** Nmap automation, PCAP forensics, TLS protocol analysis, web and API vulnerability analysis, reverse engineering, YARA-based malware triage, cryptographic weakness detection, cloud IAM and S3 security, CTF authoring
- **Embedded and Systems:** ESP32, custom firmware, BLE, wifi and radio signal work, Jetson edge deployment, concurrency and multithreading, memory management, IPC, process scheduling
- **Core Computer Science:** data structures, algorithms, time and space complexity analysis, object-oriented design, operating systems, computer organization, discrete mathematics, linear algebra
- **Engineering Practice:** spec-driven development, test-driven development, mutation testing, CI gating and release automation, performance profiling, schema-level data-integrity debugging, technical writing and documentation